

EXECUTIVE MEMO / FOR DECISION

Data Center Concerns on Aircraft Approach

July 29, 2026 • Transform Airports AI Council

AI-assisted decision support. The accountable human owner must verify the sources, local fit, and any action.

To: MWAA Executive Leadership **From:** Jennifer I. Kandel, Airspace/Airfield Planner, Planning Department

Bottom line

Do not advance this project at the currently proposed location. The site sits inside the RWY 1L approach and RWY 19R departure surfaces of a Category II/III ILS runway with ALSF-2 approach lighting.¹ FAA retains review authority because this project directly implicates aircraft safety and prior federal investment under Section 743.²

Why it holds

Thermal plumes. AIM § 7-6-16 names cooling-tower exhaust as a hazard producing airframe damage, aircraft upset, and possible engine failure, with turbulence "over 1,000 feet above the top of the stack or cooling tower," most critical in and around approach and departure corridors.³ FAA's 2015 Technical Guidance Memorandum classifies such plumes as "incompatible with airport operations."⁴ FAA reached the same conclusion in CEC Docket 15-AFC-01, finding that a power plant's cooling tower plumes posed a significant hazard to aircraft departing Long Beach Airport.⁵

Wildlife attractant. AC 150/5200-33C requires that stormwater management facilities near runways used by turbojet aircraft be designed so as not to create above-ground standing water.⁶ The proposed open-water retention basin, sited within the approach corridor and inside the RPZ, cannot satisfy that standard by design.

ILS integrity. FAA Order 6750.16E is explicit: "Placing an object outside the critical area does not guarantee non-interference with the ILS signal in space."⁷ Data centers combine large metallic mass, UPS and generator emissions, and thermal turbulence — three interference vectors against a Cat II/III localizer and glideslope. FAA Technical Operations owns that certification and is not a party to any lease.

Grant Assurance exposure. Assurances 19 (Operation and Maintenance), 20 (Hazard Removal and Mitigation), and 29 (Airport Layout Plan) each attach independently.⁸ Enforcement under 14 CFR Part 16 can order corrective action, withhold future AIP grants, and impose civil penalties up to three times any illegally diverted revenue.⁹

Strongest objection

The strongest counter is MWAA's own precedent. Its 2018 Western Lands transaction with Digital Realty covered 424 acres; MWAA's release said the corresponding ALP change was subject to NEPA and an Environmental Assessment was being prepared.¹⁰ That precedent shows data-center use can be considered at IAD, but not outside FAA review.

Section 743 preserves FAA jurisdiction where a project materially affects aircraft safety, ground safety, or prior federal investment.² Those tests are plainly implicated here; MWAA should assume FAA review rather than rely on the 45-day Notice of Intent window.²

What to do now

1. **Halt project advancement** pending formal coordination with the FAA Airport District Office.
2. **File FAA Form 7460-1** and require a Determination of No Hazard covering height, exterior lighting, plume dispersion, and NAVAID compatibility.¹¹
3. **Redirect the deal, not the developer.** Direct Commercial Real Estate to identify alternative parcels within the Airport Master Plan's designated non-aeronautical zones; the Western Lands envelope is a working precedent.
4. **Route Grant Assurance exposure to General Counsel.** Any ALP amendment for this parcel should re-enter the master-plan process — not a project-level review.

Sources

¹ FAA runway and instrument approach data, KIAD RWY 01L/19R (ALSF-2, Cat II/III ILS). FlightAware, flightaware.com.

² FAA Reauthorization Act of 2024, Section 743, as summarized in FAA Preliminary Instructions Memorandum on ALP Approval, October 3, 2024, faa.gov; AAAE Regulatory Alert, October 2024, alerts.aaae.org.

³ Federal Aviation Administration, Aeronautical Information Manual § 7-6-16 (current through Change 3, September 5, 2024), faraim.org.

⁴ FAA Technical Guidance Assessment Tool for Thermal Exhaust Plume Impact, September 24, 2015, faa.gov.

⁵ California Energy Commission, Docket 15-AFC-01 (Puente Power Project), FAA aeronautical study findings on cooling-tower plume hazard to departing aircraft at Long Beach Airport, efling.energy.ca.gov.

⁶ FAA Advisory Circular 150/5200-33C, Hazardous Wildlife Attractants On or Near Airports, faa.gov.

⁷ FAA Order 6750.16E, Siting Criteria for Instrument Landing Systems (April 10, 2014), faa.gov.

⁸ FAA Airport Sponsor Assurances (April 2025 revision), Assurances 19, 20, and 29, faa.gov.

⁹ 14 CFR Part 16 (Rules of Practice for Federally-Assisted Airport Enforcement), ecf.gov; civil penalty ceiling under 49 U.S.C. § 46301(a)(3).

¹⁰ Metropolitan Washington Airports Authority, "Airports Authority Announces Sale of 424 Acres of Western Lands at Dulles International," 2018, mwaa.com.

¹¹ FAA Obstruction Evaluation / Airport Airspace Analysis process, 14 CFR Part 77 and FAA Form 7460-1, ocaaa.faa.gov.